

Thesis Submission Approval Meeting

Speaker Notes

Self-Adaptive Model-Based Protection for Future Power Systems

Anoop V Eluvathingal · Department of Electrical Engineering

IIT Madras · June 2026

How to use these notes: These notes are structured as a full speaker script for each of the 22 slides. Each section begins with the *key message* — the single takeaway the audience should retain — followed by the detailed verbal script, suggested emphasis points, and anticipated committee questions with guidance on how to handle them. Total recommended talk time: **30–35 minutes** plus 10–15 minutes Q&A. Per-slide time targets are shown in each section header.

Contents

1	Title Slide	(≈1 min)	2
2	Presentation Outline	(≈1 min)	2
3	The IBR Revolution & the Protection Crisis	(≈2.5 min)	2
4	Research Gaps & Objectives	(≈2 min)	3
5	SAMBP Framework: Three-Tier Architecture	(≈2.5 min)	4
6	Unified Inverse-Estimation Framework	(≈2 min)	4
7	IBR Characterisation & Failure Modes	(≈2 min)	5
8	Ch 4: SAMBP-87L — Adaptive Line Differential (Method)	(≈2 min)	6
9	Ch 4: SAMBP-87L — Validated Results	(≈1.5 min)	6
10	Ch 5: SAMBP-SyncOC — Generator Over-Current	(≈2 min)	7
11	Ch 6: SAMBP-87T — Transformer Differential with SPRT	(≈1.5 min)	7
12	Ch 7: Adaptive Distance Protection	(≈1.5 min)	8
13	Ch 8: Microgrid Mode-Switching Protection	(≈1.5 min)	8
14	Ch 9: Non-Blocking EKF Digital Twin	(≈2 min)	9
15	Ch 10: Physics-Informed Fault Classifier	(≈2 min)	9
16	Ch 11: Wide-Area Protection Coordinator	(≈2 min)	10
17	Ch 12: Cybersecurity Architecture	(≈1.5 min)	11
18	Consolidated Performance Across All Modules	(≈1.5 min)	11
19	Conclusions & Contributions	(≈2 min)	12
20	Research Outcomes	(≈1 min)	13

21 Thesis Content & Chapter–Publication Mapping	($\approx$ 1 min) 13
22 Thank You / Questions	(Q&A) 13

Slide 1. Title Slide

(≈1 min)

Key message – Introduce yourself, your thesis title, and your guide. Establish the seriousness and scope of the problem in one sentence.

Script: “Good morning. I am Anoop V Eluvathingal, Roll Number EE18D203, registered for the Doctor of Philosophy programme in the Department of Electrical Engineering at IIT Madras. My thesis is titled ‘*Self-Adaptive Model-Based Protection for Future Power Systems*’, and I am working under the guidance of Dr K Shanti Swarup.

This thesis addresses a fundamental crisis that is unfolding in power grids worldwide: the relay protection systems that have kept our networks safe for decades were designed for synchronous generators, and they are beginning to fail as inverter-based resources take over the grid. This work presents a complete, mathematically grounded framework to solve that problem.”

Tip: Pause briefly after “beginning to fail” to let the gravity of the problem register. Make eye contact with each committee member. Keep this slide to under 60 seconds.

Slide 2. Presentation Outline

(≈1 min)

Key message – The thesis is organised into four coherent parts. Set expectations clearly so the committee can follow the arc.

Script: “I have organised this presentation into four parts.

Part I covers the motivation and the SAMBP framework itself: why conventional protection fails under high IBR penetration, what gaps currently exist in the literature, and the architecture of the solution I am proposing.

Part II covers five adaptive protection modules, one for each relay type studied in the thesis: the line differential relay, generator over-current, transformer differential, distance relay, and microgrid protection.

Part III covers three advanced layers built on top of those modules: a real-time digital twin using an Extended Kalman Filter, a physics-informed neural network for fault classification, and a wide-area protection coordinator.

Part IV concludes with the cybersecurity layer, a consolidated results comparison, and a summary of contributions and research outcomes.

I will spend the most time on Parts I through III, where the novel contributions are concentrated.”

Tip: Use the left-hand column of the slide as a natural anchor for ‘Part I and II’ and the right-hand column for ‘Part III and IV’. Do not read the bullets; speak from memory.

Slide 3. The IBR Revolution & the Protection Crisis

(≈2.5 min)

Key message – Five concrete failure mechanisms show that the IBR transition is not an incremental change — it breaks the physical assumptions baked into every existing protection standard.

Script: “Let me begin with the problem. The global transition to inverter-based resources — solar PV, wind turbines, battery storage — is accelerating. Under the Indian government’s 2030 renewable targets, the fraction of generation from IBRs will exceed sixty percent on

many substations. I define a dimensionless parameter, k_{ibr} , as the ratio of IBR apparent admittance to the total bus admittance. A fully synchronous grid is $k_{ibr} = 0$; a fully IBR grid is $k_{ibr} = 1$.

The table on the slide shows what happens to the available fault current as k_{ibr} rises. A synchronous machine contributes six times rated current during a fault. A grid-following inverter, by design, limits its output to 1.0 to 1.2 times rated current to protect its semiconductor switches. At $k_{ibr} = 0.6$, the total fault current is less than two per unit.

This creates five specific failure mechanisms for conventional protection, which I have listed on the left. First, the differential current in an 87L relay may never exceed the fixed slope setting when k_{ibr} is low, causing a missed trip. Second, when k_{ibr} is high and there is CT measurement error, the relay may false-trip on load current. Third, distance relays over-reach because IBR reactive injection shifts the apparent impedance locus. Fourth, IBR control-mode transitions during faults produce angle excursions that fool directional elements. Fifth, in islanded microgrids, current reverses direction during mode transitions, making directional overcurrent protection unreliable.

The critical point is that all five failures arise from a single root cause: the relay has a fixed setting designed for $k_{ibr} = 0$ and no mechanism to track the true, time-varying k_{ibr} of the network.”

Anticipated question: “What is the basis for the k_{ibr} definition?” — k_{ibr} is the ratio of the Norton equivalent admittance of IBR sources to total bus admittance in the positive-sequence equivalent circuit. It reduces to the generation-mix fraction under balanced conditions and is directly measurable from PMU phasor data.

Slide 4. Research Gaps & Objectives

(≈2 min)

Key message – Six specific gaps in the literature, and six matching objectives that this thesis addresses one-to-one.

Script: “Existing work in the literature addresses IBR protection in a piecemeal fashion: a correction factor for differential relays here, a harmonic-blocking improvement there. What is missing is a *unified, physics-aware adaptive framework* that applies across all relay types, updates in real time, and is provably stable.

I have identified six specific gaps. First, no relay standard or product accounts for real-time k_{ibr} as a variable setting input. Second, the protection implications of grid-forming IBR control modes — as opposed to grid-following — are largely unstudied. Third, there is no inverse-estimation framework that is simultaneously applicable to differential, impedance, and overcurrent relays. Fourth, digital-twin approaches proposed in the literature lack bounded, non-blocking update laws, meaning a slow convergence can delay the protection decision. Fifth, wide-area protection systems ignore IBR-induced coherency breakdown, which invalidates classical swing-group assumptions. Sixth, there is no adversarially robust ML layer integrated with the protection relay chain.

Each of these gaps maps directly to one of my six objectives, which are listed on the right. Together, they define the scope of the SAMBP framework.”

Tip: You do not need to read every objective. After listing the gaps, say “each of these maps directly to an objective, which I will demonstrate as we go through the chapters.”

Anticipated question: “Why is it necessary to solve all six simultaneously?” — The thesis shows that solving any subset in isolation still leaves the overall protection system vulnerable. For example, an accurate EKF oracle is useless if the ML classifier is susceptible to adversarial attack. The strength of the SAMBP architecture is precisely its cross-layer coherence.

Slide 5. SAMBP Framework: Three-Tier Architecture

(≈2.5 min)

Key message – SAMBP is a *stack*, not a collection of independent fixes. The EKF oracle feeds all modules; the WAPC integrates their outputs; the security layer hardens the entire chain.

Script: “The SAMBP framework is organised into three tiers, as shown in the diagram on this slide.

Tier 1 is the Parameter Oracle. This is the EKF-based digital twin described in Chapter 9. It runs continuously on PMU phasor data and maintains real-time estimates of the three key latent parameters: $\hat{k}_{ibr}$, the line impedance $\hat{Z}_{line}$, and the IBR admittance $\hat{Y}_{ibr}$. These estimates are made available to all modules over a low-latency local bus.

Tier 2 is the collection of adaptive protection modules. Each module — 87L, SyncOC, 87T, distance, and microgrid — queries the oracle at every update cycle and adjusts its operating thresholds accordingly via what I call the bounded update law. Critically, each module also incorporates a Stage-2 confidence gate: before issuing a trip command, the module computes a confidence score γ based on the uncertainty in $\hat{k}_{ibr}$, and only trips if γ exceeds a configurable threshold of 0.85. This eliminates false trips caused by transient estimation noise.

Tier 3 is the Wide-Area Protection Coordinator. It receives trip candidates from all zone relays, maintains a real-time topology graph, and applies an IBR-aware coherency model to determine the minimum-cost fault isolation action.

Below all three tiers sits the cybersecurity layer, which authenticates every IED message and adversarially hardens the PINN classifier.

The key architectural principle is that the Stage-2 confidence gate decouples the speed of the EKF oracle from the correctness of the trip decision. Even if the oracle is still converging after a topology change, the relay will hold its last trusted estimate rather than issuing a potentially incorrect trip.”

Anticipated question: “What happens if the EKF oracle fails?” — Each module has a safe fallback: if the oracle has not provided an update within a configurable timeout (default 3 cycles), the module reverts to its nominal fixed-parameter setting and the Stage-2 gate forces the confidence score to zero, effectively disabling adaptive operation until the oracle recovers. This is the fail-secure behaviour described in Chapter 9.

Slide 6. Unified Inverse-Estimation Framework

(≈2 min)

Key message – All SAMBP modules share a single Levenberg–Marquardt inverse-estimation core. Convergence is proved analytically; the bounded update law is the key stability ingredient.

Script: “Chapter 2 establishes the mathematical foundation shared by every module in the framework.

The central problem is: given a stream of phasor measurements z_t from IEDs and PMUs, estimate the latent protection-relevant parameters $\theta = \{k_{ibr}, Z_{line}, Y_{ibr}\}$ in real time. This is an inverse problem, and I solve it using the Levenberg–Marquardt algorithm, which is a damped Gauss–Newton method. The update rule is shown on the left side of the slide. The Jacobian J captures how the relay model output changes with each parameter; the damping term λI regularises the inverse and controls the step size.

The bounded update law is the modification I introduce. The learning rate α_t is not constant; it is multiplied by a factor $(1 - k_{ibr}^2)$ which smoothly drives the update rate to zero as the network approaches fully IBR operation. This prevents the estimator from taking large steps in uncertain territory and is the mechanism that enables the Lyapunov stability proof in Theorem 2.1.

The Stage-2 confidence gate is computed as a Gaussian function of the distance between the current estimate $\hat{\theta}$ and the last locally-validated operating point θ^* . The threshold $\gamma_{min} = 0.85$ was chosen to give a false-trip probability below 10^{-4} under CT noise levels consistent with IEC 60044.

The five methods listed in the table on the right are not independent tools. They are layered: LM provides the parameter estimates; EKF tracks their dynamics; SPRT tests hypotheses against those estimates; PINN provides a physics-consistent feature space; and projected gradient enforces coordination constraints.”

Anticipated question: “Is LM not computationally expensive for real-time use?” — The parameter vector is only 3–5 dimensional. A single LM iteration involves a 5-by-5 matrix inversion, which takes under 50 microseconds on standard IED hardware. The EKF runs in a separate thread at 20 ms intervals; the LM correction is applied asynchronously. Full timing budgets are detailed in Appendix C.

Slide 7. IBR Characterisation & Failure Modes

(≈2 min)

Key message – Chapter 3 derives the quantitative conditions for each failure mode and establishes k_{ibr} as the single parameter that predicts them all.

Script: “Chapter 3 provides the analytical foundation for the failure mode analysis. I develop a generalised IBR fault current model: the IBR contribution during a fault is k_{ibr} times the rated current, at a control angle ϕ_{ctrl} that is set by the LVRT strategy and is not directly observable by the relay.

The three failure modes are shown in the table. At low k_{ibr} , below about 0.2, the differential current I_{diff} never reaches the fixed slope boundary — the relay sees a fault that looks like normal load unbalance and restrains. At high k_{ibr} combined with CT measurement error, the load current at full IBR output can exceed the slope boundary and trigger a false trip. The third mode is impedance overreach: IBR reactive injection adds a component to the apparent impedance that pushes the locus outside the Zone 1 boundary even for an in-zone fault.

The insight from this analysis is that all three modes are characterised by a different regime of k_{ibr} . A relay that knows k_{ibr} can adapt its settings to remain in the correct operating region across the entire range. This is precisely what the adaptive slope law in Chapter 4 achieves.”

Tip: Point to the table values as you describe each failure mode. Emphasise that this is

a *unified* characterisation — not three separate problems but one problem viewed from three angles.

Slide 8. Ch 4: SAMBP-87L — Adaptive Line Differential (Method) (≈ 2 min)

Key message – The adaptive slope law is a smooth, monotonically decreasing function of k_{ibr} ; it has a closed-form Lyapunov stability proof.

Script: “Chapter 4 applies the SAMBP framework to the line differential relay, Type 87L.

The adaptive slope law is the core contribution. The conventional relay uses a fixed slope $k_m = 0.30$. I replace this with a function of the estimated k_{ibr} : $k_m(k_{ibr}) = k_{base} \cdot (1 - k_{ibr})^2 / (1 + k_{ibr})^2 + k_{floor}$.

This function has a natural interpretation. The numerator $(1 - k_{ibr})^2$ shrinks as IBR contribution grows, reducing the slope and keeping the relay sensitive. The denominator $(1 + k_{ibr})^2$ prevents the slope from ever reaching zero: as $k_{ibr} \rightarrow 1$, k_m approaches k_{floor} , a small positive value that guards against measurement noise.

The stability proof uses the Lyapunov function $V = \|k_{ibr} - \hat{k}_{ibr}\|^2$. Under bounded CT noise and the bounded update law, $\dot{V} < 0$, which guarantees that the estimated k_{ibr} converges to a neighbourhood of the true value. The radius of that neighbourhood is of order σ_{noise} , the CT noise standard deviation.

The relay logic involves six steps. Phasors are sampled at 1 kHz; k_{ibr} is estimated every 20 ms via the LM update; the adaptive slope is recomputed; the operate/restrain criterion is evaluated; the Stage-2 confidence gate is checked; and the trip command is issued only if both stages pass.”

Anticipated question: “How is the relay stable if k_{ibr} itself is uncertain?” — The Lyapunov proof shows stability in the *error* dynamics, not in the absolute value. Even if k_{ibr} lags the true k_{ibr} by up to one per-unit standard deviation, the slope law still keeps the differential threshold within the safe operating envelope. The Stage-2 gate provides an additional margin.

Slide 9. Ch 4: SAMBP-87L — Validated Results

(≈ 1.5 min)

Key message – Perfect TPR of 1.000 and zero false trips across 110 fault scenarios, including full-IBR grid conditions where fixed relays fail in nearly one in six cases.

Script: “I will now present the key quantitative results for the 87L module.

The test matrix covers 110 scenarios: eleven values of k_{ibr} from 0 to 1.0, five fault types (AG, BG, CG, BCG, ABC), two CT error levels (0% and 5%), and two network loading conditions. The baseline is a fixed-slope relay with $k_m = 0.30$.

The headline result is: SAMBP-87L achieves a True Positive Rate of 1.000 and a False Positive Rate of 0.000. The fixed relay achieves a TPR of 0.783 and a FPR of 14.8%. The missed trips occur entirely in the $k_{ibr} > 0.6$ region; the false trips occur in the $k_{ibr} > 0.4$ plus CT error region. SAMBP eliminates both failure modes simultaneously.

The average trip time is 21 ms for SAMBP versus 22 ms for the fixed relay. The Stage-2 gate adds approximately one cycle of additional decision latency, which is acceptable under IEC 60255-27 requirements for Class P protection systems.

The EKF oracle achieves a k_{ibr} RMSE of 0.024 pu across all scenarios, which is consistent

with the theoretical convergence bound from Chapter 2.”

Tip: The committee will focus on the TPR/FPR numbers. Be prepared to explain: (a) what constitutes a “scenario” in the test matrix, (b) whether the network model is representative of real Indian grid conditions, and (c) how the results would change under higher CT saturation.

Anticipated question: “Is a 4-bus test network sufficient to validate the approach?” — The 4-bus network was chosen to provide an analytically tractable reference for the theoretical bounds. A 14-bus IEEE test network simulation (described briefly in Appendix B) shows comparable TPR gains. Hardware-in-loop validation on RTDS is planned as future work and is listed in the thesis conclusions.

Slide 10. Ch 5: SAMBP-SyncOC — Generator Over-Current (≈2 min)

Key message – Online adaptation of the IEC inverse-time pickup setting cuts spurious pickups by 58% while maintaining coordination margins across the full IBR range.

Script: “Chapter 5 applies the SAMBP framework to generator over-current protection, the relay type typically designated 50/51 in IEC 60255 terminology, and which I call SyncOC because I focus on synchronous generators operating in an IBR-rich environment.

The fundamental problem is that the classical IEC inverse-time characteristic assumes a deterministic fault current magnitude, which is set once at the commissioning stage based on the synchronous generator’s subtransient reactance. When k_{ibr} rises, the generator’s fault contribution is depressed because the IBR sources absorb a fraction of the fault current. The relay sees less current but its pickup setting has not changed, which can cause a spurious pickup during a heavy load event that resembles a fault at high k_{ibr} .

My solution adapts the pickup current I_{set} as a function of $\hat{k}_{ibr}$, using the same LM estimator as the 87L module. The coordination constraint — that the primary relay trips before the backup, with a coordination time interval CTI between them — is enforced at every update step via a projected gradient step that keeps the solution within the feasible coordination set.

The results show a 58% reduction in spurious pickups, from 38 to 16 events per 100-hour simulation, and an R^2 of greater than 0.999 between the adaptive curve and the theoretical IEC characteristic. The coordination margin improves from 0.82 to 0.94 because the adaptive pickup tracks the actual fault current more faithfully.”

Anticipated question: “Does the projected gradient step always converge within one update cycle?” — In all 1,200 tested operating points it converged within 3 iterations, which takes under 0.1 ms. The theoretical convergence condition is that the gradient of the coordination constraint is locally Lipschitz, which holds for the IEC inverse curves.

Slide 11. Ch 6: SAMBP-87T — Transformer Differential with SPRT (≈1.5 min)

Key message – Replacing 2nd-harmonic blocking with an SPRT that has an IBR-updated likelihood model halves the decision window and reduces IBR-induced false positives by 90%.

Script: “Chapter 6 deals with transformer differential protection, relay type 87T. The classical approach to blocking false trips caused by magnetising inrush current is to measure

the second harmonic component of the differential current. If the second harmonic exceeds 15–20% of the fundamental, the relay restrains. This works well for synchronous-machine-fed transformers, but Type-4 IBR inverters produce waveforms that may have significant second-harmonic content even during genuine internal faults, and conversely, they can produce inrush-like fundamental distortion that lacks second harmonic content. The 2nd-harmonic criterion therefore becomes unreliable.

I replace it with the Wald Sequential Probability Ratio Test. The SPRT accumulates a log-likelihood ratio Λ_k at each sample. When Λ_k crosses an upper threshold A , a trip is issued; when it crosses a lower threshold B , the relay restrains. The thresholds are derived analytically from the desired Type I and Type II error probabilities, both set to 1% per IEC 60255 requirements.

The key SAMBP contribution is that the H_1 fault likelihood model is updated in real time using the IBR harmonic injection spectrum from $\hat{Y}_{ibr}$. This keeps the SPRT calibrated to the actual network waveform environment.

The results: inrush rejection improves from 94.1% to 99.6%, internal fault TPR improves from 96.8% to 99.1%, IBR-induced false positives fall from 8.7% to 0.9%, and the mean decision sample count halves from 14 to 7, meaning the relay trips faster.”

Slide 12. Ch 7: Adaptive Distance Protection

(≈1.5 min)

Key message – Zone 1 overreach drops from 34% to 4% of events by compensating the IBR-induced impedance shift in real time, at less than 1 ms speed penalty.

Script: “Chapter 7 addresses distance protection, relay type 21. The problem here is the impedance overreach caused by IBR reactive injection.

During a fault on a line with a high-IBR source at the remote end, the IBR injects reactive current that adds a component ΔZ_{ibr} to the apparent impedance seen at the relay. This shifts the impedance locus radially outward on the R-X plane. If the shift is large enough, the locus falls outside the Zone 1 boundary even though the fault is genuinely inside Zone 1.

My solution computes the compensation term $\hat{\Delta Z}_{ibr}(k_{ibr})$ using the LM estimator output and adds it to the Zone 1 boundary every 10 ms. The adapted boundary moves outward to meet the shifted locus, restoring correct Zone 1 coverage.

For severe IBR conditions where $k_{ibr} > 0.5$, SAMBP additionally enables Zone 2 acceleration: it temporarily allows an instantaneous Zone 2 trip to prevent unacceptable fault-clearance time.

The results: correct Zone 1 trips improve from 74% to 96%, overreach events fall from 17 to 2 out of 50 test faults, and underreach is eliminated entirely. The average trip time increases by less than 1 ms due to the zone adaptation computation.”

Slide 13. Ch 8: Microgrid Mode-Switching Protection

(≈1.5 min)

Key message – A Bayesian mode detector drives a mode-adaptive directional element, eliminating the 22% mis-operation rate at the grid-to-island transition boundary.

Script: “Chapter 8 addresses protection for microgrids that operate in both grid-connected and islanded modes. The transition between modes is the most dangerous moment: the pre-fault load flow used to set directional element polarity becomes invalid within one cycle of the transition, and the fault current magnitude, direction, and impedance all

change.

My approach uses a Bayesian mode detector that computes the likelihood of each mode given the recent measurement sequence z_t and the estimated k_{ibr} . When the detector switches to islanded mode with probability above 0.95, the directional element automatically switches from the pre-fault load-flow reference to a polarity-agnostic differential criterion with an adaptive pickup at 1.05 times rated current.

The key design choice is that the mode switch is *latched*: once islanded mode is confirmed, the relay does not revert until grid reconnection is explicitly confirmed by a two-out-of-three GOOSE vote from neighbouring relays. This prevents chattering.

The results show a mode-detection accuracy of 98.7%, a transition mis-operation rate of 1.4% versus 22% for the fixed relay, and an islanded-fault TPR of 97.3% versus 71%. There are zero false island detections across 100 test scenarios.”

Slide 14. Ch 9: Non-Blocking EKF Digital Twin

(≈2 min)

Key message – The EKF digital twin achieves k_{ibr} RMSE of 0.024 pu with sub-2-cycle convergence and is non-blocking: the relay never waits for the oracle.

Script: “Chapter 9 describes Tier 1 of the SAMBP architecture — the EKF-based digital twin that acts as the parameter oracle for all other modules.

The digital twin maintains a continuous state estimate $\hat{\theta}$ using the standard EKF prediction–correction cycle. The prediction step uses a first-order Markov model for parameter evolution; the correction step uses the PMU voltage and current phasors as observations. The observation function $h(\hat{\theta})$ is the forward model of the network derived from the nodal admittance matrix, updated with the latest breaker status from the SCADA system.

The non-blocking property is the key architectural novelty. The EKF runs in a separate execution thread with a dedicated real-time priority. If, after a topology change, the EKF has not converged within 3 cycles — 60 ms at 50 Hz — the oracle signals a hold condition, and each protection module continues operating with the last validated $\hat{\theta}$. The module does not wait. This ensures that even transient oracle failures do not introduce protection delays.

The bounded update law prevents adversarial data injection from causing large parameter steps. Any correction that would change θ by more than $\delta_{\max}$ per cycle is clipped and flagged as a potential integrity anomaly.

The results show an RMSE of 0.024 pu for k_{ibr} , 0.018 pu for $\hat{Z}_{line}$, and 0.031 pu for $\hat{Y}_{shunt}$, across all operating conditions including 5% PMU measurement noise. Convergence after a topology change takes fewer than 2 cycles, and the maximum update latency is 1.8 ms.”

Anticipated question: “How does the EKF perform under bad data, for example a stuck PMU?” — The EKF residual monitor flags measurements whose innovation exceeds three standard deviations of the expected measurement noise. Flagged measurements are excluded from the correction step. If all measurements from a PMU are flagged for more than 5 consecutive cycles, the oracle issues a degraded-mode signal and modules fall back to conservative fixed settings. Full analysis is in Chapter 9, Section 4.

Slide 15. Ch 10: Physics-Informed Fault Classifier

(≈2 min)

Key message – Adding a KCL physics residual to the neural-network loss function boosts accuracy on high-IBR faults from 64% to 94% — a 46% relative gain.

Script: “Chapter 10 develops the physics-informed neural network fault classifier. Pure machine-learning approaches trained on historical fault data fail on IBR-modified waveforms because the training data distribution is dominated by synchronous-machine faults. Adding a physics residual to the loss function forces the network to respect Kirchhoff’s Current Law at each bus, which is the constraint that all fault types share regardless of the source type.

The PINN loss has two terms. The first is the standard cross-entropy classification loss on labelled fault data. The second is the KCL residual: the squared norm of $\mathbf{K}\hat{\mathbf{V}} - \hat{\mathbf{I}}$, where $\mathbf{K}$ is the nodal admittance matrix and $\hat{\mathbf{V}}, \hat{\mathbf{I}}$ are the voltage and current phasors predicted by the network for the current input. The weight $\lambda_{\text{phys}} = 0.3$ was chosen by cross-validation to balance the two terms.

The input feature vector includes eight elements: the three phase current magnitudes and angles, the zero-sequence and negative-sequence components, and crucially, the oracle’s k_{ibr} and ϕ_{ctrl} estimates. Including the oracle outputs as features is what allows the PINN to generalise across the full k_{ibr} range.

The results: overall accuracy rises from 81.4% for a standard DNN to 96.1% for the PINN. For high-IBR conditions with $k_{\text{ibr}} > 0.6$, the gain is from 64.2% to 93.8%. Physical consistency — meaning the predicted fault state satisfies KCL to within a tolerance of 10^{-3} per unit — rises from 51.2% to 99.7%.”

Anticipated question: “How was the PINN training dataset constructed?” — The training set comprises 15,000 simulated fault scenarios generated using the parameterised IBR fault model from Chapter 3, with k_{ibr} uniformly sampled from $[0, 1]$, five fault types, three fault impedance levels, and three loading conditions. Details and the dataset are available on IEEE DataPort (reference given in the thesis).

Slide 16. Ch 11: Wide-Area Protection Coordinator

(≈2 min)

Key message – Replacing electromechanical swing-group coherency with an IBR-aware coherency matrix improves fault localisation from 71% to 94% and halves backup clearance time.

Script: “Chapter 11 describes the Wide-Area Protection Coordinator at Tier 3 of the SAMBP stack.

Classical wide-area protection systems group generators into coherent swing groups using electromechanical swing equations. In a network with 60% or more IBR, those swing groups are invalid: inverters do not participate in electromechanical oscillations. The coherency grouping therefore assigns incorrect backup priorities.

My replacement is the IBR coherency matrix $\mathbf{C}(k_{\text{ibr}})$, which groups buses based on their combined IBR admittance vector similarity rather than inertia-weighted swing angles. The matrix is computed from the oracle’s k_{ibr} values at each bus and is updated every 100 ms, which is fast enough to track topology changes but slow enough not to overload the communication system.

The WAPC server receives trip candidates from all zone relays over IEC 61850 GOOSE messages. It builds a topology graph from real-time breaker status, applies the coherency matrix to identify the fault island, and solves an Integer Linear Programme to find the minimum-cost set of breaker operations that isolates the fault while retaining maximum load.

The ILP has a time limit of 50 ms per decision. In all tested scenarios it solved within 35 ms. For complex topologies with more than 20 candidate breakers, a greedy warm-start is used which finds a near-optimal solution within 10 ms.

The results: fault localisation accuracy rises from 71.2% to 94.5%, unnecessary trips fall from 18.4% to 4.1%, load preserved at the end of the isolation sequence rises from 63% to 87.6%, and backup fault clearance time is halved from 420 ms to 210 ms.”

Anticipated question: “What is the communication latency assumed?” — The model assumes a worst-case one-way GOOSE latency of 4 ms per hop, consistent with IEC 61850 Performance Class P2. Sensitivity analysis in Chapter 11, Section 5, shows that the fault-localisation accuracy remains above 90% up to a latency of 12 ms, beyond which the ILP topology snapshot becomes stale.

Slide 17. Ch 12: Cybersecurity Architecture

(≈1.5 min)

Key message – HMAC authentication and adversarial PINN training reduce the composite risk index by 54.9% with less than 0.3 ms overhead — within IEC 62351 latency budgets.

Script: “Chapter 12 addresses the cybersecurity of the SAMBP architecture. As the system relies on real-time PMU data and IED messages, it presents two attack surfaces that I identify and defend against.

The first threat is data integrity attack. An adversary who can modify PMU measurements in transit can inject false $\hat{k}_{ibr}$ values and induce the protection system to mal-operate. The defence is HMAC-SHA256 authentication on every IED message, using keys managed by the IEC 62351-8 key management infrastructure. A relay that receives an unauthenticated or tampered message discards it and holds the last trusted parameter estimate. The authentication overhead is 0.3 ms per message, which is within the IEC 62351 latency budget for Class P2 systems.

The second threat is adversarial machine learning attack. An adversary who knows the PINN architecture can craft a perturbation $\delta\mathbf{x}$ with magnitude less than 5% of the input range that causes the classifier to mis-identify a fault. Without defence, this drops the PINN accuracy from 96% to under 52%.

The defence is adversarial training using Fast Gradient Sign Method and Projected Gradient Descent examples. The retrained network achieves a certified robustness radius of 0.04 pu, meaning no perturbation within that radius can change the classification. Accuracy under attack recovers to 91.4%.

The composite risk index — a weighted sum of attack success rate, impact severity, and relay latency — falls from 1.000 to 0.451, a 54.9% reduction.”

Slide 18. Consolidated Performance Across All Modules

(≈1.5 min)

Key message – Every SAMBP module outperforms its fixed-parameter baseline, and the gains are consistent across the full IBR penetration range $k_{ibr} \in [0, 1]$.

Script: “Let me now bring together the results from all chapters into a single comparison table.

Looking across the rows, every SAMBP module shows a clear improvement over its fixed-parameter baseline. The 87L relay goes from 78% TPR to 100% and eliminates all false trips. The SyncOC relay cuts spurious pickups by 58%. The 87T relay reduces IBR-

induced false positives by 90%. The distance relay improves correct Zone 1 discrimination from 74% to 96%. The microgrid relay cuts transition mis-operations from 22% to 1.4%. The PINN classifier improves from 81% to 96% overall accuracy. The WAPC improves fault localisation from 71% to 94%. And the cybersecurity layer cuts the composite risk index nearly in half.

A key aspect of these results is that they are measured under the *same* test conditions as the baselines, with no tuning advantage given to SAMBP. The only difference is the adaptive parameter update. This validates the core claim of the thesis: that real-time knowledge of k_{ibr} is sufficient to repair the protection failures caused by the IBR transition, without requiring changes to relay hardware or communication infrastructure.”

Tip: Spend about 30 seconds on this slide — the committee will read the table themselves. Focus on the narrative: one cause (unknown k_{ibr}), one solution (adaptive estimation), consistent gains across all relay types.

Slide 19. Conclusions & Contributions

(≈2 min)

Key message – Six principal contributions, each novel by a specific technical criterion; four directions for future work that connect directly to real-world deployment.

Script: “Let me summarise the principal contributions of this thesis.

First, the SAMBP framework itself. To my knowledge, this is the first unified inverse-estimation architecture that applies a single parameter oracle to multiple relay types simultaneously. Prior work has proposed adaptive corrections for individual relay types in isolation; this thesis provides the theoretical and implementation framework that unifies them.

Second, the adaptive 87L result. A TPR of 1.000 with zero false trips across the full IBR range is a result that no prior adaptive differential relay paper has reported. The key enabling technology is the Stage-2 confidence gate, which I believe is also a novel contribution to IED trip-chain design.

Third, the non-blocking EKF digital twin. The non-blocking property — where the relay never waits for the oracle — is a specific design decision that, again to my knowledge, has not been formalised in prior digital-twin protection literature.

Fourth, the PINN fault classifier. The use of a KCL physics residual as a loss-function term is established in the scientific ML literature, but this is its first application to power-system protection classification under IBR conditions.

Fifth, the IBR-aware WAPC. The substitution of the IBR coherency matrix for electromechanical swing groups is a conceptually clean contribution that directly addresses why wide-area protection systems are degrading in practice.

Sixth, the integrated cybersecurity layer. Combining HMAC authentication with adversarial PINN training as a unified security architecture for a protection stack is a novel design.

For future work, the three most important directions are: RTDS hardware-in-loop validation, field deployment on a representative portion of the TANGEDCO network in Tamil Nadu, and extension of the IBR model to grid-forming inverters with virtual inertia, which introduce dynamics not captured by the current k_{ibr} formulation.”

Slide 20. Research Outcomes

(≈1 min)

Key message — Four journal papers, two conference papers, one patent, one codebase, two datasets — all derived directly from the thesis chapters.

Script: “The research in this thesis has produced the following outcomes.

Four journal papers are under review. The 87L work is submitted to IEEE Transactions on Power Delivery. The SyncOC work is submitted to IET Generation, Transmission and Distribution. The PINN classifier paper is under review at IEEE Transactions on Neural Networks and Learning Systems. The WAPC paper is being prepared for Electric Power Systems Research.

Two conference papers have been published: the EKF digital-twin work was presented at IEEE PES ISGT-Europe 2025, and the adversarial robustness paper was presented at IEEE TENCON 2025.

A patent application has been filed through IIT Madras on the adaptive relay system with real-time IBR parameter estimation.

The SAMBP simulation framework is approximately 4,200 lines of Python and MATLAB code and is available on request. Two fault-signature datasets for IBR-grid conditions have been uploaded to IEEE DataPort and are publicly accessible.”

Tip: Keep this brief. If a committee member asks for a specific paper, say the preprint is available and offer to share it after the meeting.

Slide 21. Thesis Content & Chapter–Publication Mapping (≈1 min)

Key message — The thesis is 13 chapters plus four appendices, approximately 204 pages, with clear traceability from chapters to publications.

Script: “This slide shows the complete structure of the thesis. All thirteen chapters and four appendices are listed with estimated page counts and their associated publication outputs.

The thesis begins with an introduction and mathematical foundations in Chapters 1 and 2. Chapters 3 through 12 are the research chapters, each corresponding to one module or cross-cutting concern of the SAMBP framework. Chapter 13 contains the conclusions and future-work directions.

The four appendices contain the proof details from Chapter 2, the full system data for all test networks, the pseudocode listings for all algorithms, and additional data tables for the SyncOC coordination study.

The total estimated length is approximately 204 pages, which is consistent with IIT Madras PhD dissertation norms for Electrical Engineering.”

Slide 22. Thank You / Questions (Q&A)

Key message — Leave the committee with the three headline numbers: TPR=1.000, 96.1% accuracy, –54.9% risk reduction.

Script: “Thank you for your attention. To summarise in three numbers: the adaptive line differential relay achieves a true positive rate of 1.000 — no missed trips, no false trips. The physics-informed fault classifier achieves 96.1% overall accuracy with 93.8% on high-IBR scenarios where conventional deep networks fall to 64%. And the cybersecurity architecture reduces the composite risk index by 54.9% with less than 0.3 milliseconds of

authentication overhead.

I am happy to take your questions.”

Tip: Stand near the screen and maintain eye contact with the committee. Do not start listing additional results unless a committee member specifically asks. If you do not know an answer, say so directly: “That is a good question; I have not analysed that case and it would be an interesting direction.” Confidence under uncertainty is more reassuring to a doctoral committee than a speculative answer.

Frequently anticipated Q&A topics (prepare these in advance):

1. How does the method scale to a 400-bus transmission network?
The LM estimator is per-feeder; the EKF operates on the local equivalent circuit. The WAPC ILP complexity scales with the number of candidate isolation breakers, not total bus count. Full scalability analysis is in Chapter 11, Section 6.
2. What is the minimum PMU density required?
One PMU per protected feeder terminal is sufficient for the EKF. The WAPC requires at least one PMU per coherency group. Sensitivity to PMU dropout is analysed in Chapter 9.
3. How does SAMBP interact with existing distance relay firmware?
The adaptive zone boundaries are output as IEC 61850 XSWI setting messages to the existing IED. The relay firmware is unchanged. This is a key practical design constraint of the architecture.
4. Has the method been tested on real utility data?
Not yet. All results are based on high-fidelity simulation using validated IBR models. RTDS validation and TANGEDCO field trials are listed as future work.
5. What is the cost of deploying the SAMBP oracle on existing RTU hardware?
The EKF oracle requires approximately 50 MIPS of compute, consistent with mid-range IED processors (e.g., Siemens SIPROTEC 5 class). The codebase is written in C-compatible Python and has been profiled on ARM Cortex-A53.